

# Qubentoo GNU/Linux — Full Build Report

Full Gentoo overlay and bootstrap system for Qubes OS on OpenRC

Qubentoo Project

2026-04-26

Qubentoo GNU/Linux — Build Report

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## Qubentoo GNU/Linux — Build Report

**Date:** 2026-04-26

**Overlay root:** ~/qubentoo-overlay/

**Total files created:** 47

**Phases completed:** 7 (initial) + 4 porting fixes + 1 bonus doc

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### Executive Summary

Qubentoo is a complete rebase of Qubes OS onto a Gentoo Linux base using OpenRC instead of systemd. This report documents all work completed across two build sessions: the initial 7-phase scaffold and the subsequent porting issue resolution pass.

All packages use EAPI=8, the python-single-r1 eclass for Python components, and a custom qubes.eclass that strips systemd units and auto-installs OpenRC init scripts. No systemd units are present anywhere in the overlay.

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# Session 1 — Initial Scaffold (Phases 1-7)

## Phase 1 — Overlay Scaffold

| File                            | Purpose                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata/layout.conf            | masters = gentoo, repo-name = qubentoo                                                                                                                    |
| profiles/base/eapi              | EAPI=8                                                                                                                                                    |
| profiles/base/<br>make.defaults | Hardened USE flags, -systemd openrc<br>xen, Python 3.11, GRUB Xen<br>platforms                                                                            |
| eclass/qubes.eclass             | Shared helpers: qubes_src_uri,<br>qubes_src_unpack, qubes_src_install;<br>auto-strips systemd dirs; auto-installs<br>OpenRC scripts from FILESDIR/openrc/ |

**Design decision:** The eclass handles the systemd→OpenRC conversion automatically. Any ebuild that calls `qubes_src_install` will have `/lib/systemd` and `/usr/lib/systemd` removed from `${D}` and will have OpenRC scripts placed from `${FILESDIR}/openrc/` via `newinitd`.

## Phase 2 — Xen Ebuilds

**app-emulation/xen-4.17.5** — Xen 4.17.5 hypervisor, dom0 build only. No stubdom, no ioemu. Configures both `xen/` and `tools/` subdirectories.

**app-emulation/xen-tools-4.17.5** — Userspace tools with four OpenRC init scripts:

| Service    | Runlevel | Depends on             |
|------------|----------|------------------------|
| xenstored  | sysinit  | localmount             |
| xencommons | sysinit  | xenstored              |
| xenconsole | default  | xencommons             |
| xendomains | default  | xencommons, xenconsole |

`xenstored` mounts `xenfs` at `/proc/xen` and starts the `xenstore` daemon. `xendomains` saves running VMs on shutdown and restores them on next boot, respecting `XENDOMAINS_AUTO` configs in `/etc/xen/auto/`.

## Phase 3 — Core Qubes dom0 Ebuilds

Six packages covering the Qubes dom0 management stack:

| Package                         | Version | OpenRC Service              | Runlevel |
|---------------------------------|---------|-----------------------------|----------|
| sys-apps/qubes-db               | 4.2.0   | qubes-db                    | default  |
| sys-apps/qubes-core-vchan-xen   | 4.2.0   | <i>(library, no daemon)</i> | —        |
| sys-apps/qubes-libvchan         | 4.2.0   | <i>(library)</i>            | —        |
| sys-apps/qubes-core-admin       | 4.2.0   | qubesd                      | default  |
| sys-apps/qubes-core-admin-linux | 4.2.0   | qubes-core                  | default  |
| sys-apps/qubes-rpc-proxy        | 4.2.0   | qubes-rpc-proxy             | default  |

**Porting note:** qubes-rpc-proxy upstream source is qubes-core-qrexec; QUBES\_REPO is overridden in the ebuild accordingly.

qubes-libvchan is split from qubes-core-vchan-xen to allow packages that need only the library API to avoid depending on the full vchan transport daemon.

## Phase 4 — GUI Stack Ebuilds

**gui-daemon/qubes-gui-common-4.2.0** — Shared GUI protocol headers and libraries. Installed in both dom0 and domU templates.

**gui-daemon/qubes-gui-daemon-4.2.0** — dom0 X11 GUI daemon. OpenRC service waits for the X11 display using a xdpinfo poll loop before starting. DISPLAY and XAUTHORITY are configurable via /etc/conf.d/qubes-gui-daemon.

## Phase 5 — Kernel Config Fragment

kernel/qubentoo-dom0.config — A real defconfig fragment (using =y/=m/ # ... is not set syntax) covering:

**Xen dom0 support:** - CONFIG\_XEN=y, CONFIG\_XEN\_DOM0=y, CONFIG\_XEN\_PRIVILEGED\_GUEST=y - Full PCI passthrough: CONFIG\_XEN\_PCIDEV\_BACKEND=m, CONFIG\_PCI\_XEN=y - IOMMU: CONFIG\_INTEL\_IOMMU=y, CONFIG\_AMD\_IOMMU=y, CONFIG\_IRQ\_REMAP=y - Xenstore, grant tables, event channels, balloon memory

**Hardened security options:** - CONFIG\_HARDENED\_USERCOPY=y — prevents kernel slab leaks - CONFIG\_FORTIFY\_SOURCE=y — compile-time string safety - CONFIG\_CC\_STACKPROTECTOR\_STRONG=y -

CONFIG\_RANDOMIZE\_BASE=y, CONFIG\_RANDOMIZE\_MEMORY=y — KASLR -  
CONFIG\_PAGE\_TABLE\_ISOLATION=y — Meltdown mitigation -  
CONFIG\_RETPOLINE=y — Spectre v2 mitigation -  
CONFIG\_SECURITY\_LOCKDOWN\_LSM=y — kernel lockdown

**GRSecurity-equivalent (no patch): -**

CONFIG\_SLAB\_FREELIST\_RANDOM=y, CONFIG\_SLAB\_FREELIST\_HARDENED=y -  
CONFIG\_SECURITY\_YAMA=y — ptrace restriction -  
CONFIG\_INIT\_ON\_ALLOC\_DEFAULT\_ON=y, CONFIG\_INIT\_ON\_FREE\_DEFAULT\_ON=y  
- CONFIG\_MODULE\_SIG=y, CONFIG\_MODULE\_SIG\_FORCE=y - #  
**CONFIG\_MODULE\_FORCE\_UNLOAD is not set** — as required

Apply the fragment with:

```
scripts/kconfig/merge_config.sh .config qubentoo-dom0.config
```

---

## Phase 6 — Bootstrap Script

scripts/bootstrap-dom0.sh — Idempotent 11-step dom0 bootstrap:

1. Prerequisite check (emerge, eselect, portageq)
2. Set hardened Gentoo profile
3. Install eselect-repository, register qubentoo overlay
4. Write dom0 USE/CFLAGS/MAKEOPTS to /etc/portage/make.conf
5. Write package.accept\_keywords for all ~amd64 packages
6. Update @world
7. Emerge Xen hypervisor then tools
8. Emerge all Qubes packages in dependency order
9. Configure OpenRC runlevels (sysinit + default) 9b.  
**Interactive DM prompt** — sets QUBES\_GUI\_DM in conf.d (see Issue 4)
10. Configure GRUB for Xen multiboot (EFI or MBR fallback)
11. Merge kernel config fragment

Every step checks whether it has already run before acting, making re-runs safe.

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## Phase 7 — Gentoo Template VM Script

scripts/build-gentoo-template.sh — Builds a Qubes-compatible template tarball from scratch:

1. Downloads latest hardened-openrc stage3 automatically
2. Creates a raw .img disk image (configurable size, default 20 GB)
3. Partitions with parted (single ext4 root), attaches via loopback
4. Extracts stage3, bind-mounts /proc, /sys, /dev for chroot
5. Configures portage inside chroot (hardened profile, USE flags)
6. Updates @world

7. Installs Qubes guest packages via the qubentoo overlay
8. Writes /etc/qubes/guid.conf
9. Enables qubes-db in OpenRC default runlevel
10. Sets hostname, locale (en\_US.UTF-8), PV-compatible fstab
11. Packages as root.img + template.conf tarball

Install the result with:

```
qvm-template install --local /var/lib/qubes/vm-templates/gentoo-  
qubentoo-1.tar  
qvm-create --template gentoo-qubentoo --label green my-gentoo-vm
```

---

## Session 2 — Porting Issue Resolution

### Issue 1 — Guest Agent Ebuilds

Two new ebuilds for the domU (template/AppVM) side:

**sys-apps/qubes-core-agent-4.2.0** - Source: [github.com/QubesOS/qubes-core-agent-linux](https://github.com/QubesOS/qubes-core-agent-linux) - Installs: /usr/lib/qubes/, /etc/qubes/, udev rules  
- Two OpenRC services: - qubes-agent (default): starts qubesdb-daemon, waits for socket - qubes-qrexec-agent (default, after qubes-agent): qrexec daemon - Depends on: qubes-db, qubes-libvchan, qubes-rpc-proxy, dev-python/qubesdb, nftables, dbus

**gui-daemon/qubes-gui-agent-4.2.0** - Source: [github.com/QubesOS/qubes-gui-agent-linux](https://github.com/QubesOS/qubes-gui-agent-linux) - Installs: qubes-gui binary, /etc/qubes/guid.conf - OpenRC qubes-gui-agent: depends on qubes-agent, qubes-qrexec-agent, and the configurable display manager (see Issue 4)

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### Issue 2 — Manifest Generation and QA Scripts

**scripts/gen-manifests.sh** - Finds all .ebuild files in the overlay tree  
- Skips packages with RESTRICT="mirror" (no distfiles to hash) -  
Runs ebuild <path>.ebuild manifest for each - Logs to manifest-gen.log with per-package pass/fail status - Prints a summary table and exits non-zero if any manifest failed

**scripts/verify-overlay.sh** - Runs repoman scan from the overlay root  
- Saves full output to repoman-report.txt - Parses output for error/warning counts (handles both line-count and summary-line formats) - Exits non-zero on any errors; --strict flag also fails on warnings - Requires: emerge dev-util/repoman

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## Issue 3 — Python Binding Stub Ebuilds

**dev-python/qubesdb-4.2.0** - Source: [github.com/QubesOS/qubes-core-qubesdb](https://github.com/QubesOS/qubes-core-qubesdb) (python/ subdir) - Builds the qubesdb C extension linking against libqubesdb - Uses python-single-r1, supports python3\_{10,11,12} - RDEPEND: sys-apps/qubes-db

**dev-python/xen-4.17.5** - Source: Xen upstream tarball (tools/python/ subdir) - Installs xen.lowlevel, xen.xl, xenstore Python modules - Also installs pygrub (Xen bootloader helper) from tools/pygrub/ - Uses python-single-r1, supports python3\_{10,11,12} - RDEPEND: app-emulation/xen-tools - Packaged separately from xen-tools for dependency graph clarity: packages that need only the Python API don't pull in the full daemon set

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## Issue 4 — Configurable Display Manager Dependency

The display manager dependency in both qubes-gui-daemon (dom0) and qubes-gui-agent (domU) is now fully configurable via conf.d:

```
# /etc/conf.d/qubes-gui-agent
# Common values: xdm, lightdm, sddm, gdm
QUBES_GUI_DM="xdm"
```

The OpenRC init script reads this variable at source time and declares need "\${QUBES\_GUI\_DM}" in its depend() function.

**bootstrap-dom0.sh interactive prompt** (step 9b):

```
Which display manager will you use in dom0?
Options: xdm lightdm sddm gdm
Press Enter to accept default [xdm]:
```

Non-interactive use: `DM_CHOICE=lightdm bash scripts/bootstrap-dom0.sh`

Full switching instructions, USE flags, and troubleshooting are documented in [docs/display-manager.md](#).

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## Bonus — Dependency Graph

[docs/dependency-graph.md](#) contains:

- ASCII tree of the full dom0 package dependency graph
- ASCII tree of the full domU/template dependency graph
- OpenRC startup order for both dom0 and domU
- Mermaid diagram with colour-coded node types (dom0 / domU / shared / Python)

- Numbered build order lists matching bootstrap-dom0.sh and build-gentoo-template.sh
- 

## Complete File Manifest

```

qubentoo-overlay/
├── eclass/
│   └── qubes.eclass
├── metadata/
│   └── layout.conf
├── profiles/base/
│   ├── eapi
│   └── make.defaults
├── app-emulation/
│   ├── xen/
│   │   └── xen-4.17.5.ebuild
│   └── xen-tools/
│       ├── xen-tools-4.17.5.ebuild
│       └── files/
│           ├── openrc/
│           │   ├── xenstored
│           │   ├── xencommons
│           │   └── xenconsole
│           └── xendomains
│               └── confd/
│                   ├── xencommons
│                   └── xendomains
├── sys-apps/
│   ├── qubes-db/
│   │   ├── qubes-db-4.2.0.ebuild
│   │   └── files/openrc/qubes-db
│   │       └── files/confd/qubes-db
│   ├── qubes-core-vchan-xen/
│   │   └── qubes-core-vchan-xen-4.2.0.ebuild
│   ├── qubes-libvchan/
│   │   └── qubes-libvchan-4.2.0.ebuild
│   ├── qubes-core-admin/
│   │   ├── qubes-core-admin-4.2.0.ebuild
│   │   └── files/openrc/qubesd
│   │       └── files/confd/qubesd
│   ├── qubes-core-admin-linux/
│   │   ├── qubes-core-admin-linux-4.2.0.ebuild
│   │   └── files/openrc/qubes-core
│   ├── qubes-rpc-proxy/
│   │   ├── qubes-rpc-proxy-4.2.0.ebuild
│   │   └── files/openrc/qubes-rpc-proxy
│   │       └── files/confd/qubes-rpc-
proxy
│   └── qubes-core-agent/
│       ├── qubes-core-agent-4.2.0.ebuild
│       └── files/openrc/qubes-agent
│           ├── qubes-qrexec-agent
│           └── files/confd/qubes-agent
└── gui-daemon/

```

← NEW (Issue 1)

```

|   |   | qubes-gui-common/
|   |   |   | qubes-gui-common-4.2.0.ebuild
|   |   | qubes-gui-daemon/
|   |   |   | qubes-gui-daemon-4.2.0.ebuild
|   |   |   | files/openrc/qubes-gui-daemon  files/confd/qubes-gui-
daemon
|   |   |   | files/guid.conf
|   |   |   | qubes-gui-agent/ ← NEW (Issue 1
+ 4)
|   |   |   | qubes-gui-agent-4.2.0.ebuild
|   |   |   | files/openrc/qubes-gui-agent  files/confd/qubes-gui-
agent
|   |   |   | files/guid.conf
|   |   | dev-python/ ← NEW (Issue
3)
|   |   |   | qubesdb/
|   |   |   |   | qubesdb-4.2.0.ebuild
|   |   |   | xen/
|   |   |   |   | xen-4.17.5.ebuild
|   |   | kernel/
|   |   |   | qubentoo-dom0.config
|   |   | scripts/
|   |   |   | bootstrap-dom0.sh (updated: DM prompt in step 9b)
|   |   |   | build-gentoo-template.sh
|   |   |   | gen-manifests.sh ← NEW (Issue 2)
|   |   |   | verify-overlay.sh ← NEW (Issue 2)
|   |   | docs/
|   |   |   | display-manager.md ← NEW (Issue 4)
|   |   |   | dependency-graph.md ← NEW (Bonus)

```

**Total: 47 files**

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## Known Remaining Work

The following items are out of scope for the current build but represent the next logical milestones:

1. **qubes-gui-agent in build-gentoo-template.sh** — the template build script currently installs only the core agent packages. qubes-gui-agent should be added once X11 support is confirmed working in the template.
2. **Manifest files** — Manifest files per package must be generated with scripts/gen-manifests.sh on a live Portage host before the overlay can be published.

3. **qubes-core-qrexec split** — upstream qubes-core-qrexec ships both dom0 and domU components in one tree. A future ebuild split should create separate sys-apps/qubes-rpc-proxy (dom0) and sys-apps/qubes-qrexec-agent (domU) with IUSE="dom0" to select.
4. **Memory ballooning / qmemman** — qubes-core-admin includes a memory balloon daemon. The OpenRC qubes-core script currently sets xenstore entries directly; a dedicated qmemman service ebuild is needed.
5. **Dispvmm template support** — disposable VMs need additional config in /etc/qubes/guid.conf and a qubes-dispvmm OpenRC service inside the template.
6. **Qubes Manager GUI** — qubes-manager (PyQt5) is the dom0 administration UI. Not yet included; requires dev-python/PyQt5 and dev-python/qubesadmin.

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*Report generated automatically by the Qubentoo build system.  
Overlay: /home/admin/qubentoo-overlay/*